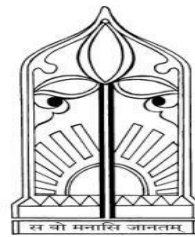


A
PROJECT REPORT
ON
“RitHan AI”
(Text To Image Generator)
SUBMITTED TO
Banaskantha District Kelvani Mandal Sanchalit
Smt. B.K.Mehta I.T.Centre (BCA College)
OPP.S.T WORKSHOP,
HIGHWAY,
PALANPUR - 385 001.



As a Partial fulfillment of the course of
BACHELOR OF COMPUTER SCIENCE & APPLICATION
Year 2025-2026

Developed by:

Exam No.

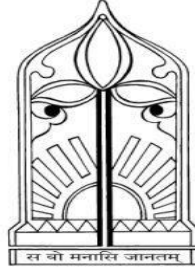
1.Farhan Khureshi
2.Ritu Modi

INTERNAL GUIDE-:

Dr. Namrata S Gupta

**Banaskantha District Kelvani Mandal Sanchalit
Smt. B.K.Mehta I.T.Centre (BCA College)OPP.S.T
WORKSHOP,
HIGHWAY,
PALANPUR - 385 001.**

Affiliated TO
(Hemchandracharya North Gujarat University, Patan)



CERTIFICATE

TO WHOM SO EVER IT MAY CONCERN

This is to certify that following student of B.C.A. have completed his project work on “RitHan AI” satisfactorily in partial fulfillment of requirement of BACHELOR OF COMPUTER SCIENCE & APPLICATION of Hemchandracharya North Gujarat University, Patan in the year 2025-26.

Serial No.	Name Of Student	Roll No.	Exam No.
1	Farhan Khureshi	70	
2	Ritu Modi	85	

Project Guide

Dr. Namrata S Gupta

Principal

Dr. Namrata S Gupta
**(Smt.B.K.Mehta I.T.Centre(BCA College),
Palanpur)**

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ABSTRACT

The **RitHan AI - Text to Image Generator** is an advanced web application designed to convert textual descriptions into unique images using artificial intelligence. This project leverages a modern tech stack including Python with the Flask framework, and a SQLite database for managing user data, history, and generated images. It features secure user authentication through Google OAuth, password management, and role-based access for both standard users and administrators.

The system's core functionality is powered by integrations with cutting-edge AI services such as the **Gemini API**, **Stability AI**, and **Hugging Face**, enabling real-time image creation. The frontend, built with HTML, Tailwind CSS, and JavaScript, provides a sleek and responsive user interface with a dynamic dashboard, community gallery, and a profile management section. The application demonstrates the practical application of AI in creative tasks, providing a seamless and efficient experience for users to generate and manage their digital art.

PROJECT PROFILE

Overview

The **RitHan** AI project is a web application designed to generate images from text descriptions. Built with Python and the Flask framework, it uses a SQLite database to manage user accounts and project data. The system automates the creative process of converting text to images, offering a powerful, efficient, and user-friendly platform.

Objectives

1. To provide a simple and intuitive interface for users to generate images from text.
2. To leverage modern AI technologies (like Gemini and Stability AI) to produce high-quality images.
3. To enable secure user management, including authentication, session handling, and password resets.
4. To create a community feature where users can explore and share their creations.
5. To ensure the application is scalable and can integrate with future cloud services.

Scope

- **User:** Can sign up, log in, generate images, manage their profile, view their generation history, and like images.
- **Admin:** Has access to an admin panel to manage users and view system data.
- **System:** Handles API integrations for image generation, manages user sessions, and stores user-generated content.
- **Frontend:** A responsive web interface with a dashboard, pricing page, and community gallery.

Future enhancements include advanced AI-driven features like text-to-image, mobile app integration, and a more robust subscription model.

ACKNOWLEDGEMENT

We, the developers of **RitHan AI**, express our sincere gratitude to **Dr. Namrata S. Gupta**, our project guide, for her invaluable guidance and unwavering encouragement throughout this project. Her expertise and insightful feedback were crucial to our success.

We would also like to thank the faculty and staff of **Smt. B.K. Mehta I.T. Centre (BCA College), Palanpur**, for their continuous support and for providing us with the resources needed to complete this work.

Finally, we are deeply grateful to our friends and family for their continuous support, inspiration, and motivation.

ABOUT INSTITUTE

Smt. B.K. Mehta I.T. Centre (BCA College), located in Palanpur, is an educational institute affiliated with **Hemchandracharya North Gujarat University, Patan**. It is managed by the Banaskantha District Kelvani Mandal. The college specializes in undergraduate programs in Computer Applications, focusing on practical, project-based learning and providing students with essential industry exposure and modern IT skills..

ABOUT SYSTEM

The **RitHan AI - Text to Image Generator** is a web-based application developed to fulfill the growing demand for AI-driven content creation. It simplifies the process of generating unique images from simple text prompts, making powerful AI tools accessible to a wide range of users.

Purpose

- Simplify the conversion of text prompts into visual art.
- Manage user accounts and session data securely.
- Provide a centralized platform for creating, viewing, and managing generated images.
- Offer role-based access for both standard users and administrators.
- Showcase a community gallery of user creations to inspire others.

Features

- Text-to-Image Generation Module:** Core functionality to create images from user prompts.
- User Management:** Secure sign-up, login, and profile management.
- Image Gallery:** A community section to display and share generated images.
- Prompt History:** Users can view and revisit their past text-to-image prompts.
- Admin Panel:** A dedicated interface for managing users and system data.
- Secure Authentication:** Implemented with Google OAuth and standard password handling.

CURRENT MANUAL SYSTEM

The manual system for creating digital art from concepts is inefficient and prone to significant challenges, such as:

- **Time-consuming creation:** Manually generating unique digital art from a concept requires significant artistic skill and can take a long time to produce.
- **Limited creativity and scope:** Without automation, the ability to rapidly iterate on different ideas or explore varied artistic styles is severely limited.
- **Difficulty in sharing and collaboration:** Sharing high-quality digital art and collaborating on projects is often cumbersome, relying on manual file transfers and external platforms.
- **Poor data security:** Without a centralized system, managing user concepts, generated images, and personal data can be disorganized and insecure.

The **RitHan AI** project automates these processes, eliminating these issues by providing a fast, secure, and systematic method for creating and managing digital art.

TECHNOLOGY USED

- **Programming Languages:** Python, JavaScript
- **Backend Framework:** Flask
- **Frontend Technologies:** HTML, Tailwind CSS, JavaScript, and custom CSS
- **Database:** SQLite
- **AI Integration:** Google's Gemini API, Stability AI API, Hugging Face
- **Libraries & Tools:**
 - **Backend:** Authlib (for Google OAuth), itsdangerous, requests, smtplib (for email), werkzeug
 - **Frontend:** Spline 3D Viewer, Swiper JS
- **Development Tools:** Visual Studio Code, Git, GitHub
- **Project Management:** Environment variables (.env) for configuration, Python virtual environment (pyvenv)

SYSTEM REQUIREMENTS

Hardware

- **Laptop/Desktop**
- **Minimum 8GB RAM** (The use of AI models like Gemini and others will be memory-intensive, making the 4GB from your example insufficient).
- **Modern CPU** (e.g., Intel i5 or higher recommended)
- **Minimum 1GB of free disk space** for project files and dependencies.

Software

- **Operating System:** Windows
- **Python:** Version 3.10 or higher
- **Web Browser:** A modern browser like Google Chrome
- **Code Editor:** Visual Studio Code
- **Version Control:** Git
- **Virtual Environment:** venv to manage project dependencies.
- **Required Python Libraries:** As specified in the requirements.txt file (e.g., Flask, Authlib, google-generativeai).

DATA DICTIONARY

Table: users

This table stores the primary information for every user in your application.

Column Name	Data Type	Description
id	int(11)	The unique ID for the user, generated automatically.
fullname	Text	The user's full name.
email	Text	The user's email address (must be unique for each user).
password_hash	Text	The user's password, stored in an encrypted format for security.
is_pro	BOOLEAN	Indicates if the user is a Pro member (True/False).
is_admin	BOOLEAN	Indicates if the user is an Admin (True/False).
credits	INTEGER	The number of image generation credits the user has remaining.
created_at	TEXT	The timestamp of when the user's account was created.
last_login_at	TEXT	The timestamp of the user's last login.

Primary Key: id

Table: generate image

This table keeps a record of every image generated by your users.

Column Name	Data Type	Description
user_id	INTEGER	The ID of the user who created the image (this links to the users table).
prompt	TEXT	The text prompt that was used to create the image.
image_url	TEXT	The path or link where the generated image is stored.
created_at	TEXT	The timestamp of when the image was created.
is_deleted	BOOLEAN	Indicates if an admin has deleted the image (True/False).

Primary Key: user_id

Table: activity log

This table is designed for admins to track user activities.

Column Name	Data Type	Description
id	INTEGER	The unique ID for the activity record.
user_id	INTEGER	The ID of the user who performed the activity (links to the users table).
action	TEXT	What the user did (e.g., 'login', 'signup', 'image_generated').
timestamp	TEXT	The timestamp of when the activity occurred.
details	TEXT	Extra information about the activity (optional).

Primary Key: id

Table: Prompt history

This table keeps a record of all prompts used by each user.

Column Name	Data Type	Description
id	INTEGER	The unique ID for the history record.
user_id	INTEGER	The ID of the user who used the prompt (links to the users table)
prompt	TEXT	The prompt text that the user submitted.
created_at	TEXT	The timestamp of when the prompt was used.

Primary Key: id

Table: liked image

This table tracks which images have been liked by which users.

Column Name	Data Type	Description
id	INTEGER	The unique ID for the 'like' action. user_ .
user_id	INTEGER	The ID of the user who liked the image (links to the users table).
image_id	TEXT	The ID of the image that was liked (links to the generated_images table).

Table: Subscriptions

This table is intended for managing payment and subscription plans in the future.

Column Name	Data Type	Description
id	INTEGER	The unique ID for the subscription.
user_id	INTEGER	The ID of the user who has the subscription (links to the users table)
plan_name	TEXT	The name of the subscription plan (e.g., "Monthly Pro").
amount	REAL	The payment amount.
payment_id	TEXT	The unique ID for the payment transaction.
start_date	TEXT	The date the subscription began.
end_date	TEXT	The date the subscription will end.

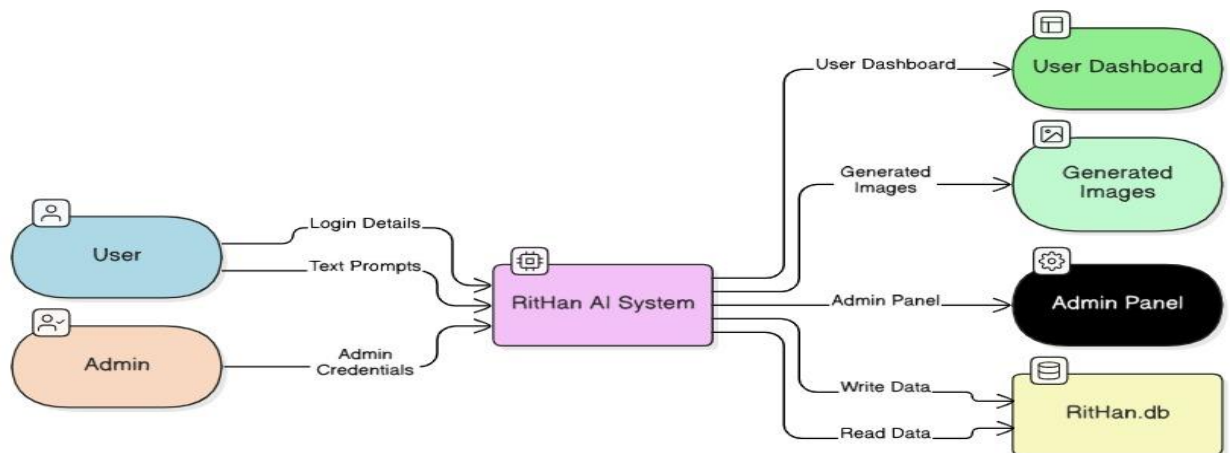
DFD DIAGRAMS

1. RitHan AI System: Context Diagram (DFD Level 0)

This first diagram provides a high-level, context-view of the entire RitHan AI System, showing its main external entities and the major data flows between them and the system itself.

Purpose

To illustrate the scope of the RitHan AI System and its direct interactions with external users, administrators, and its data repository.



❖ **Key Data Flows**

Component	Type	Description
User	External Entity	An individual interacting with the system for image generation.
Admin	External Entity	A privileged user managing the system.
RitHan AI System	System (Process)	The core application that handles all logic, data processing, and image generation.
User Dashboard	External Entity/Interface	The view or interface users see after logging in.
Generated Images	External Entity/Interface	The output interface where generated images are displayed.
Admin Panel	External Entity/Interface	The administrative interface for system management.
RitHan.db	Data Store	The persistent storage for all system data.

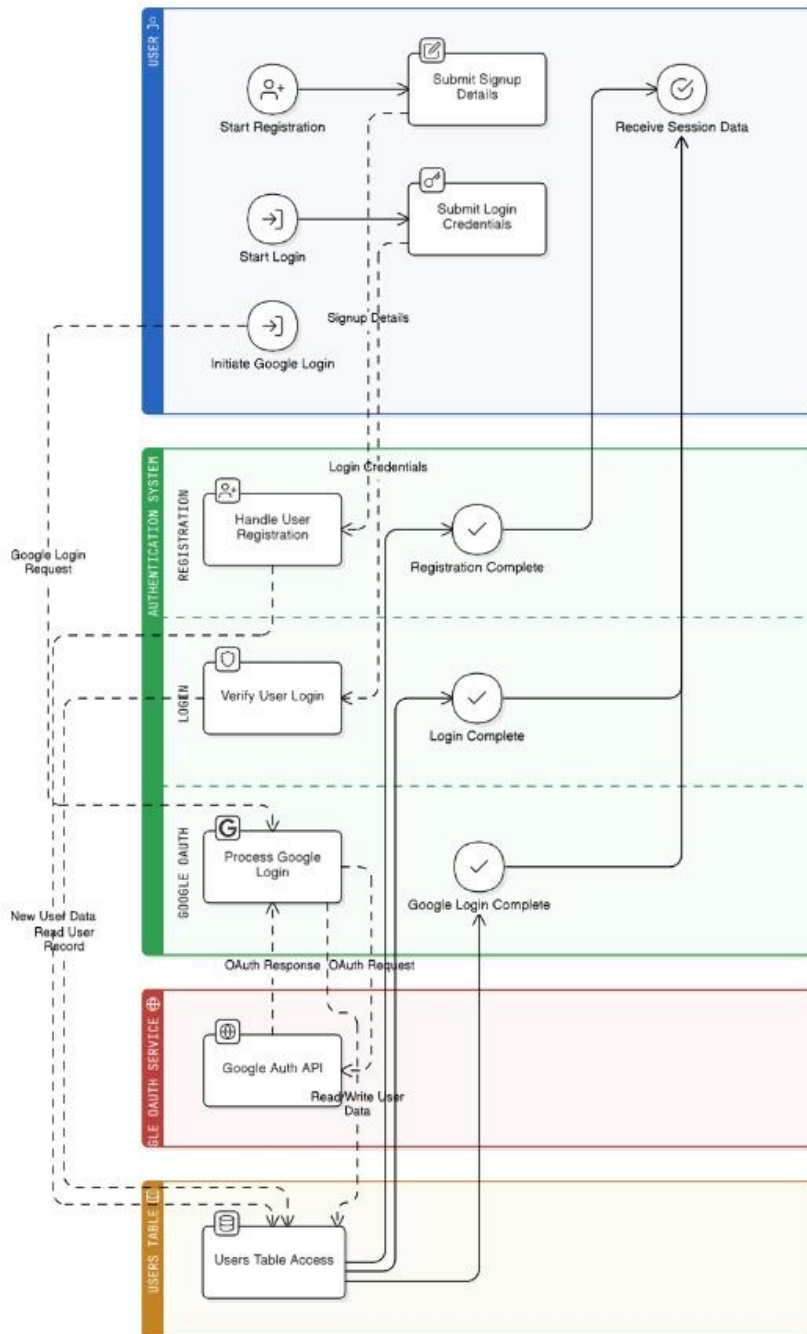
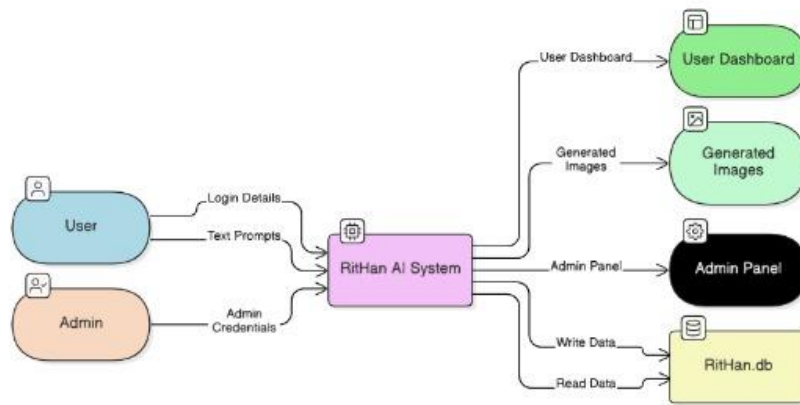
- From User to System: Login Details and Text Prompts.
- From Admin to System: Admin Credentials.
- From System to User: User Dashboard and Generated Images.
- From System to Admin: Admin Panel.
- To/From Database: Write Data (from system to DB) and Read Data (from DB to system).

2. RitHan AI System: Authentication and Authorization Flow (DFD Level 1 - Partial Decomposition)

The second diagram decomposes the initial view, specifically detailing the processes and layers involved in user authentication and login. This is often an expansion of the "RitHan AI System" process shown in the first diagram.

Purpose

To detail the various steps a user takes to register or log in, highlighting both traditional email/password and Google single sign-on (SSO) methods.



❖ Layers/Swimlanes

The diagram is structured into distinct layers to show responsibility:

1. **User UI:** Represents the user-facing forms and actions.
2. **Authentication System:** Handles core login/registration logic.
3. **Google OAuth:** Specifically manages the Google SSO process.
4. **Google Auth API Service:** Represents interaction with Google's external API.
5. **Users Table Access:** Represents the data layer (the database table for users).

❖ Key Processes

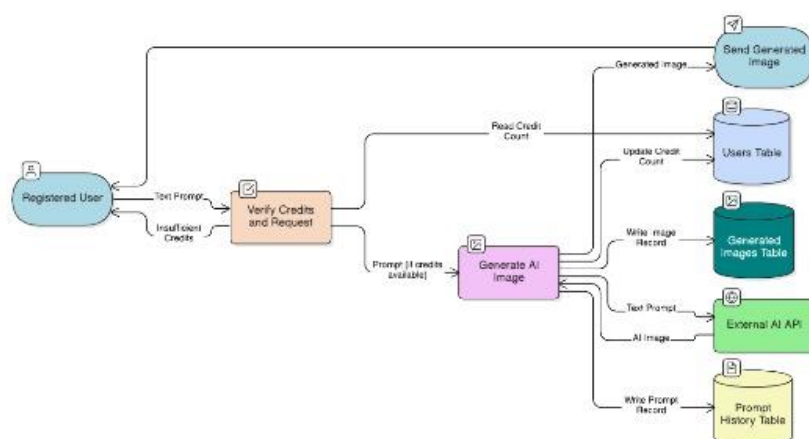
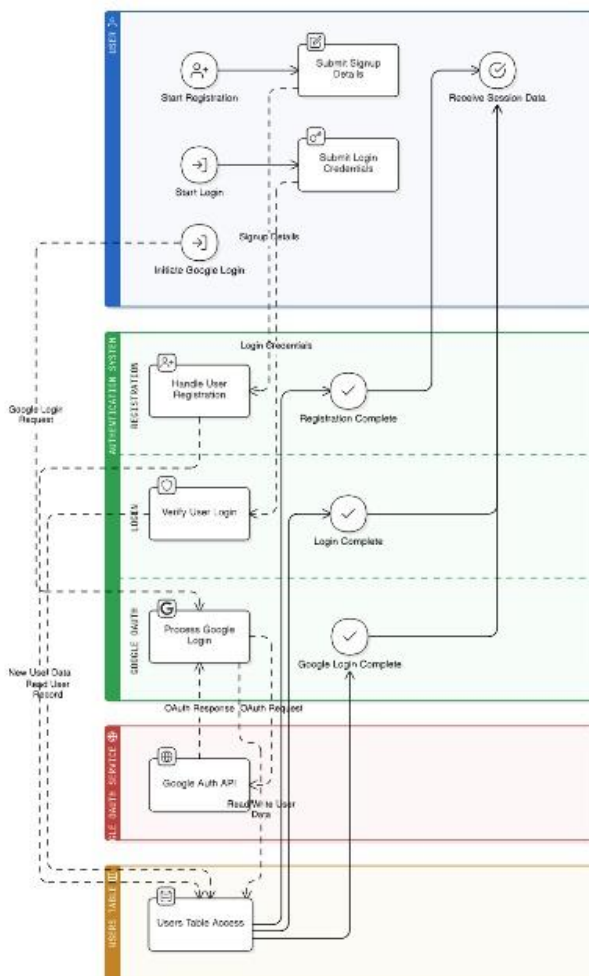
- **Traditional Login/Registration:** User initiates Start Registration or Start Login, submits details, which are processed by the **Authentication System** (Handle User Registration, Verify User Login).
- **Google Login:** User initiates Initiate Google Login. This triggers a request to the **Google Auth API Service**, which processes the OAuth response.
- **Data Flow:** All successful processes (Registration Complete, Login Complete, Google Login Complete) result in the user receiving Receive Session Data and, importantly, interact with the **Users Table Access** layer to read or write user records.

3. RitHan AI System: Image Generation Flow (DFD Level 1 - Process Decomposition)

The third diagram focuses on the core functionality of the system: image generation after a user has successfully logged in.

Purpose

To illustrate the steps, checks, and data interactions that occur from the moment a registered user submits a text prompt until the final image is generated and stored.

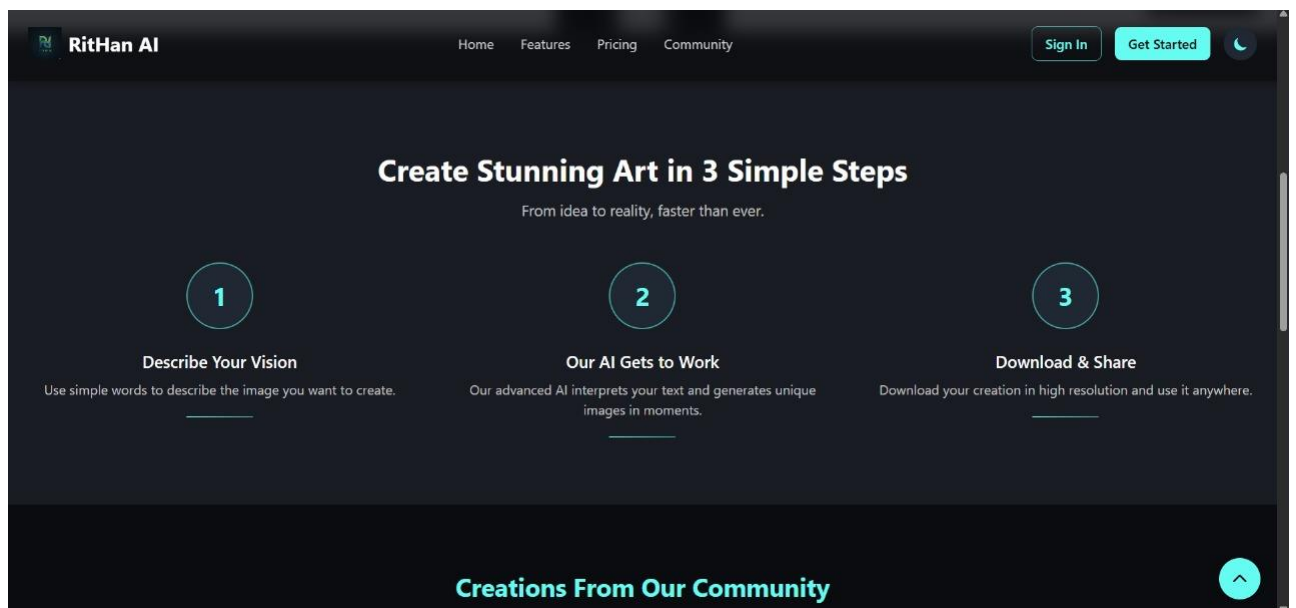
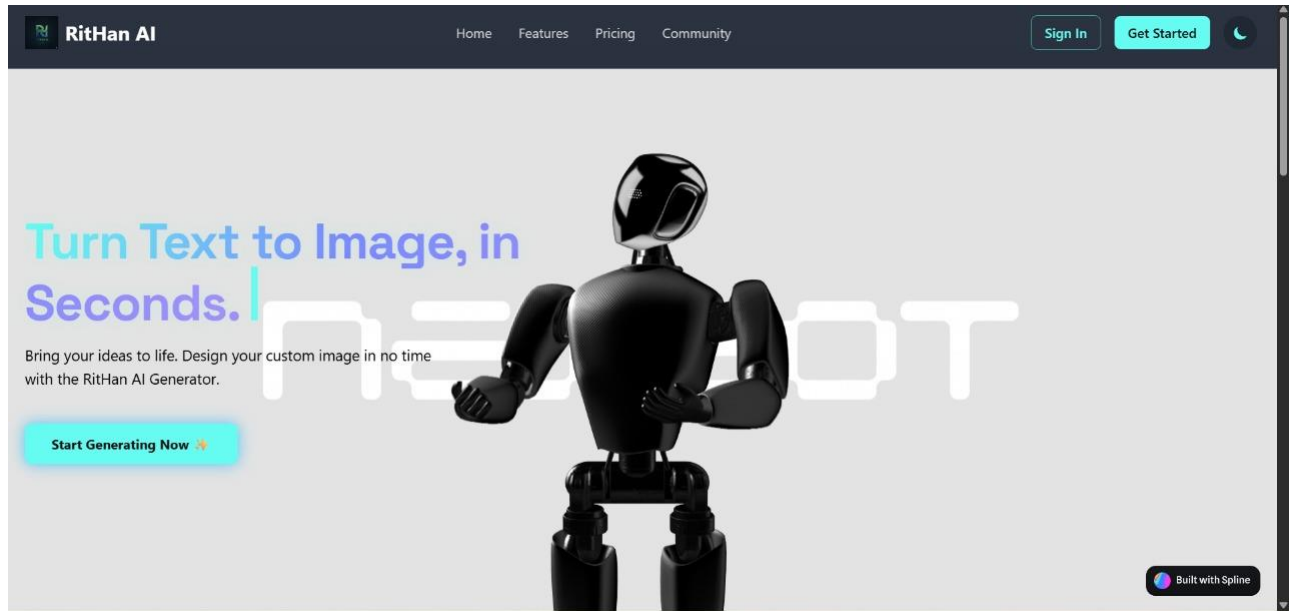


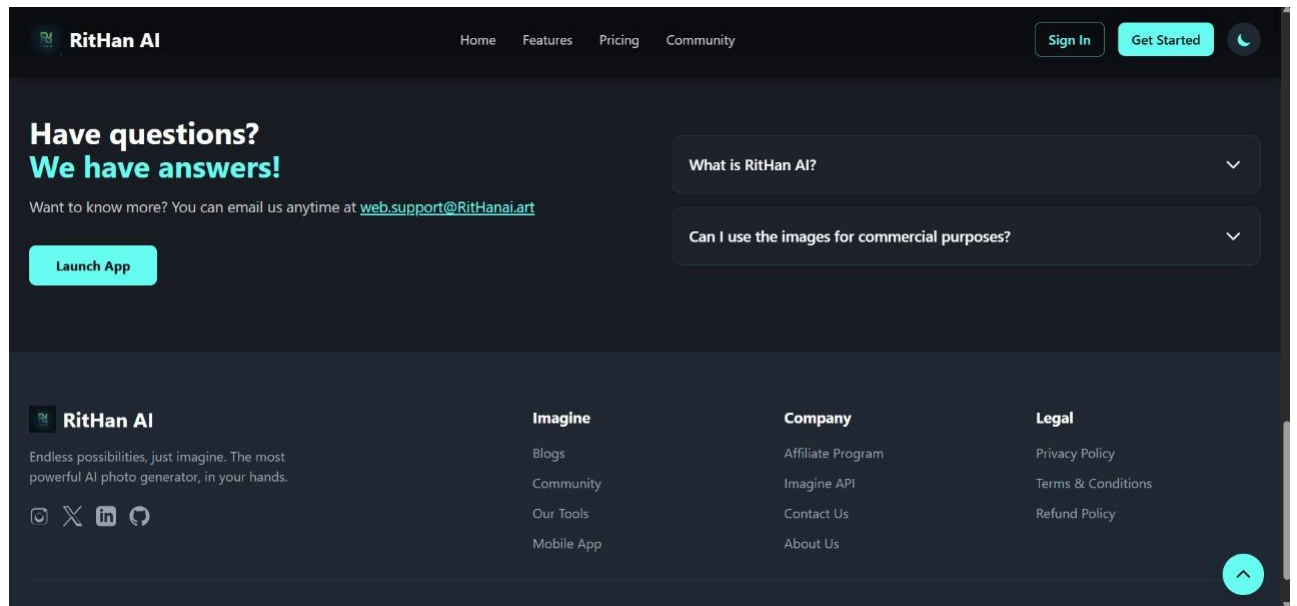
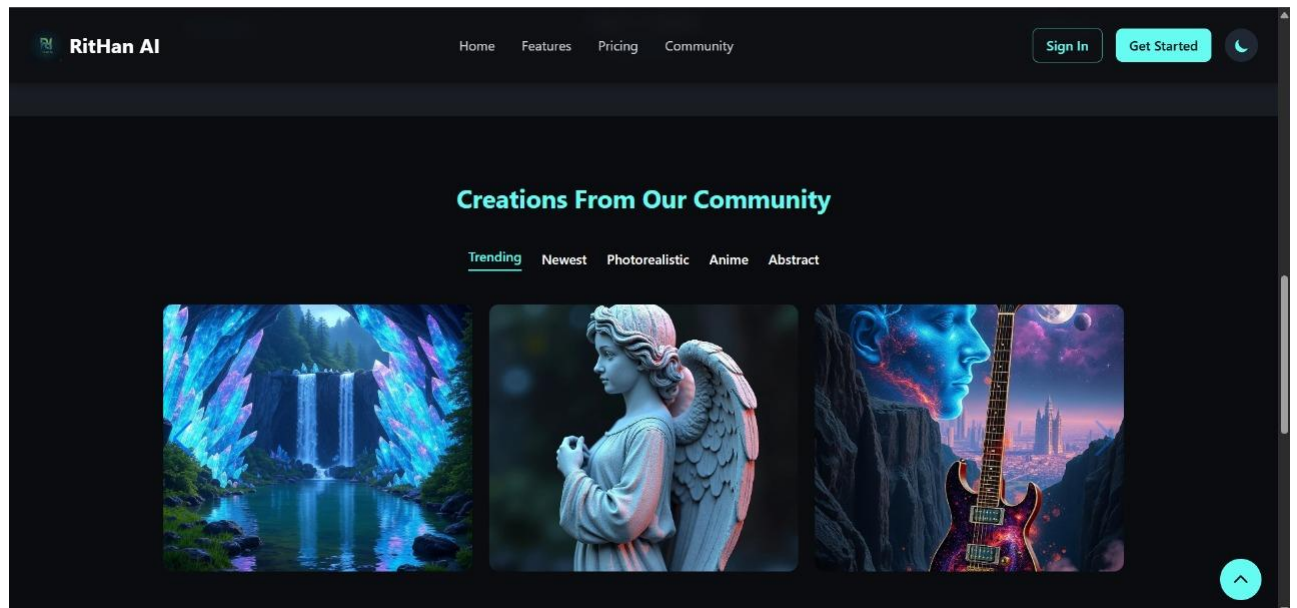
❖ Key Processes

- **1. User Request: A Registered User** submits a **Text Prompt**.
- **2. Verification:** The system runs the **Verify Credits and** Request process, which reads the user's Credit Count from the database.
 - **Failure:** If Insufficient Credits, the process terminates.
 - **Success:** If valid, the prompt continues.
- **3. Image Generation:** The request moves to the **Generate AI Image process**. This process:
 - Interacts with an **External AI API** (e.g., DALL-E, Midjourney, Stable Diffusion) by sending the Text Prompt.
 - Receives the **Generated Image** back.
- **4. Post-Generation:** Following a successful generation, the system performs critical data updates:
 - It executes Update User Count to decrement the user's credits in the **Users Table**.
 - It executes Write Image Record to store metadata and the image reference in the **Generated Images Table**.
 - It executes Write Prompt Record to store the text prompt history in the **Prompt History Table**.
- **5. Output**
 - The final Generated Image is sent to the user via the **Send Generated Image** process.

SCREEN LAYOUTS

This section is reserved for screenshots of:





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- ✓ Generate Multiple Images
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Explore the Art of AI

Discover incredible artwork created by the RitHan AI community. Get inspired, see what's possible, and start your own creative journey.

All

Trending

Photorealistic

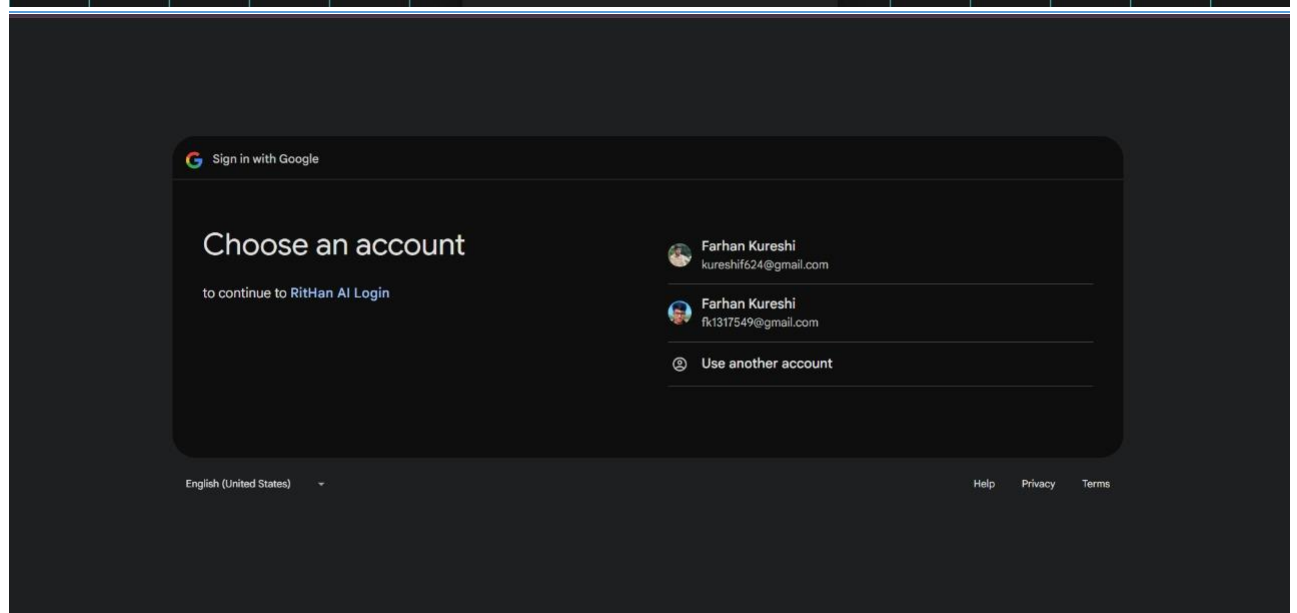
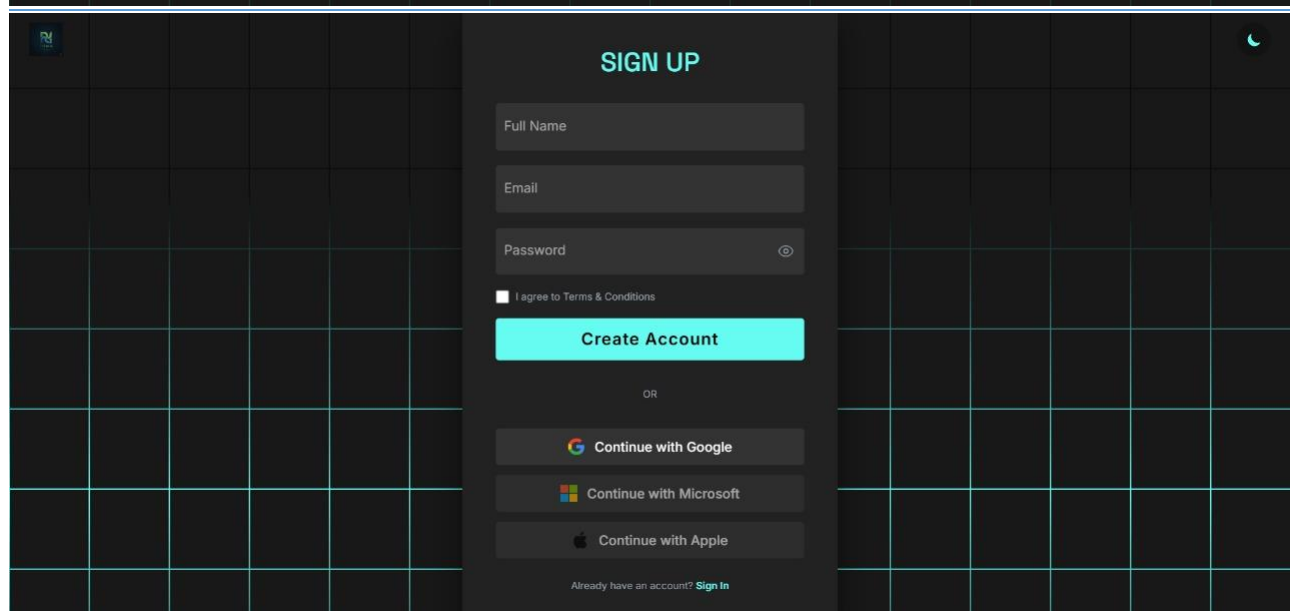
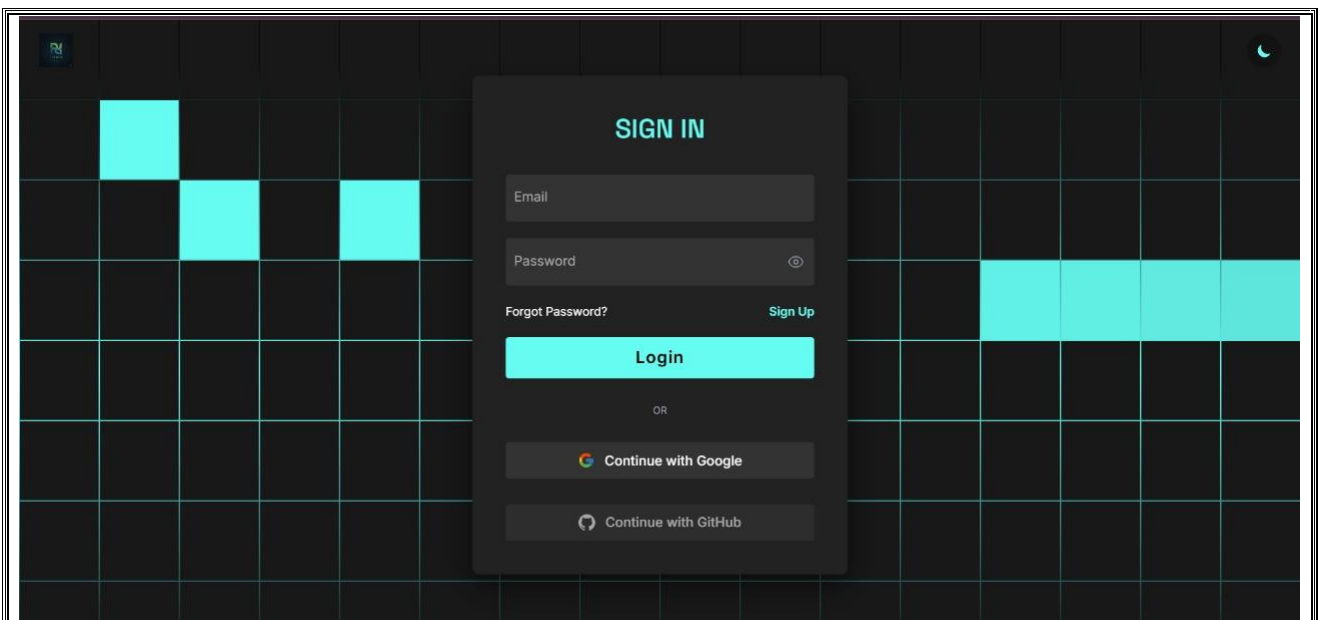
Anime

Abstract

Fantasy



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FORGOT PASSWORD

Enter your email address and we will send you a link to reset your password.

Send Reset Link

[Back to Login](#)

SET NEW PASSWORD

Reset Password

 RitHan AI

MENU

 Home

 Text to Image

 My Images

 Liked Images

 History

TOOLS

 Prompt Assistant

SETTINGS

General

Style

Image Guidance

PRO



Farhan Kureshi

Credits: Unlimited



Welcome to RitHan AI, Farhan Kureshi!

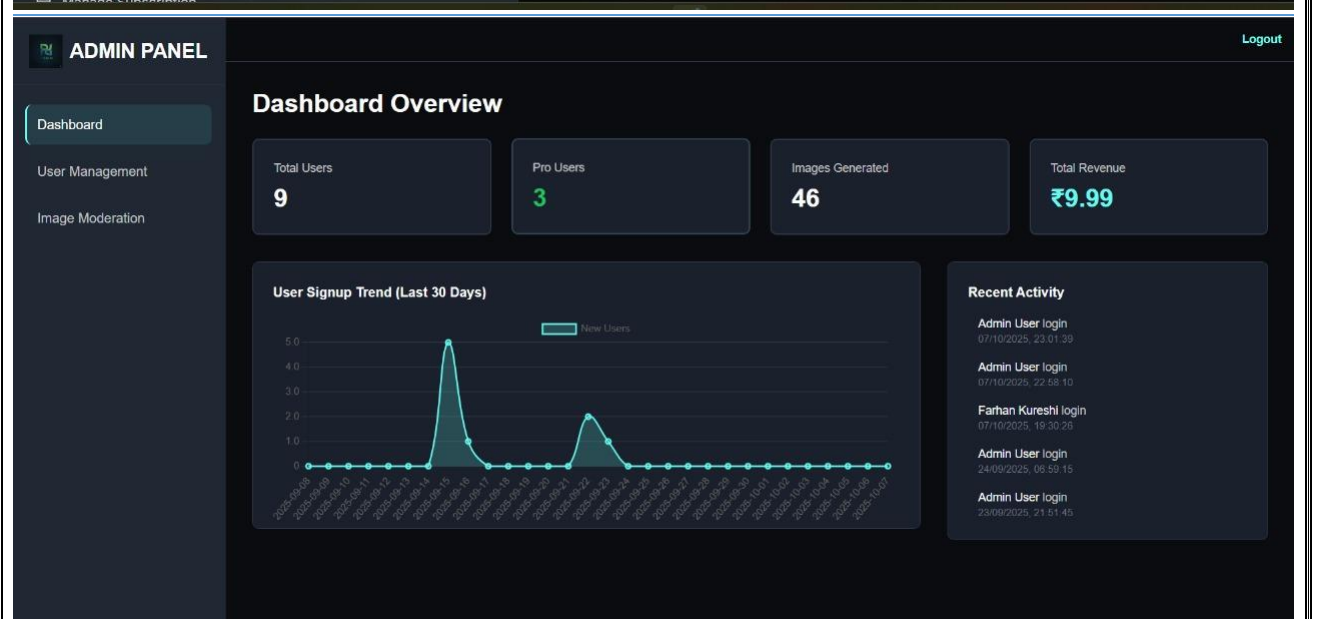
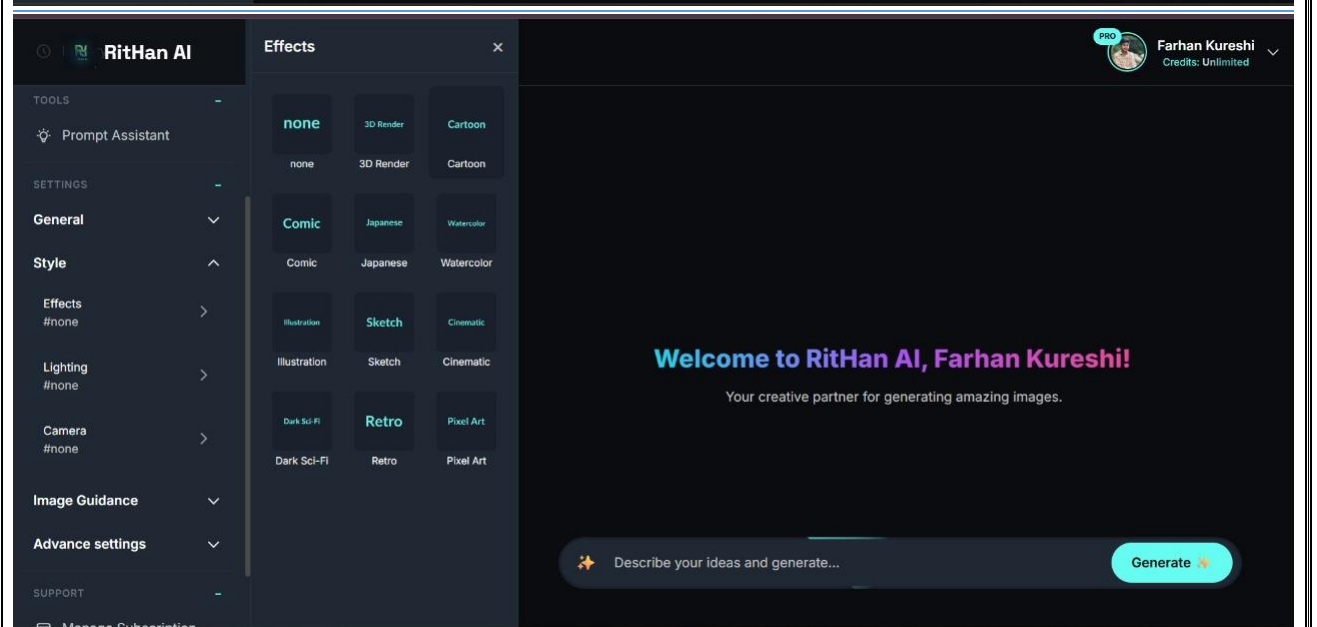
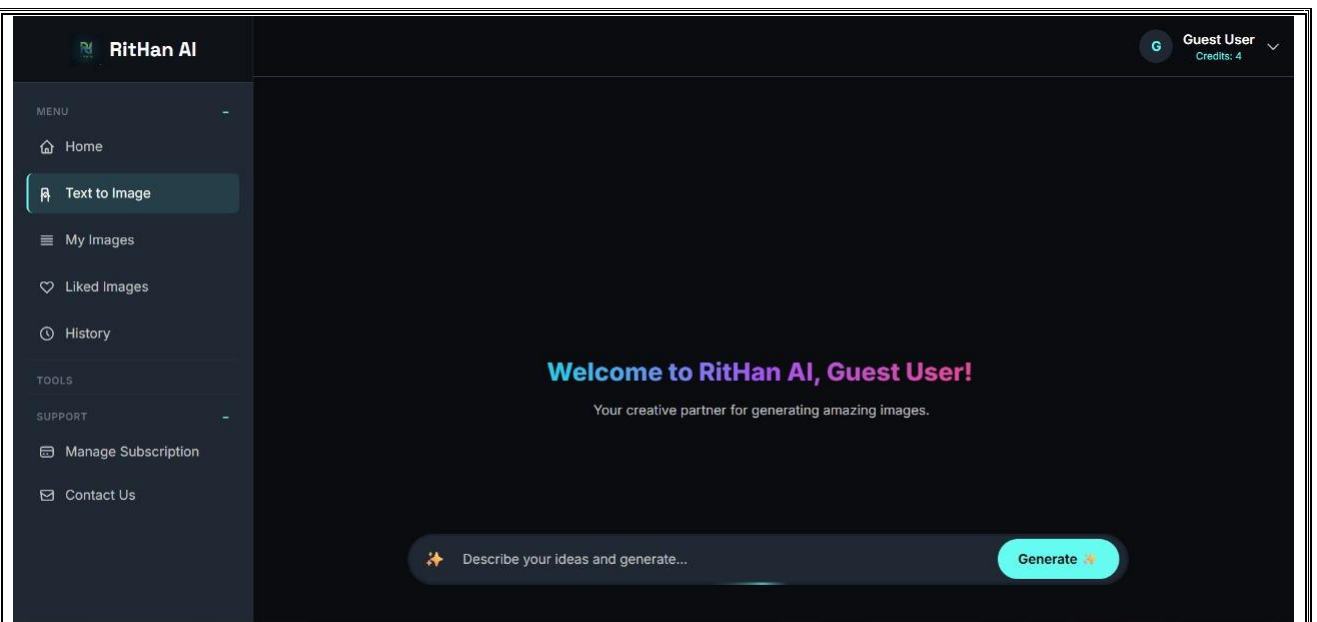
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Generate





ADMIN PANEL

Dashboard

User Management

Image Moderation

Logout

User Management

Search by name or email...

USER	STATUS	CREDITS	LAST LOGIN	ACTIONS
Mohammad mohammadkhoraiya724@gmail.com	FREE	10	Never	+ Credits Make Pro
Anish farmanq070@gmail.com	FREE	9	Never	+ Credits Make Pro
Sahil ps01022008@gmail.com	FREE	10	Never	+ Credits Make Pro
sauban kadiwalsauban18@gmail.com	FREE	109	16/09/2025	+ Credits Make Pro
Farhan Kureshi kureshi924@gmail.com	PRO	17	07/10/2025	+ Credits Make Free
Farhan Kureshi fk1317549@gmail.com	FREE	82	15/09/2025	+ Credits Make Pro
Admin User admin@rithan.ai	PRO	9999	07/10/2025	+ Credits Make Free

ADMIN PANEL

Dashboard

User Management

Image Moderation

Logout

Image Moderation

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